* [de Sá, PhD Thesis, 2010]

1.6 - Expansion of the Wakes

In practice we are mostly interested in knowing the wake functions in a region very close to the vacuum chamber's center - where the beam travels. In that region $\vec{\rho_s}$ and $\vec{\rho_w}$ are much smaller than the transverse dimensions of the chamber, allowing us to expand W in its transverse coordinates and keep only low-order terms:

$$W(\vec{\rho_s}, \vec{\rho_w}, z) \cong W_0 + \vec{\rho_s}^T \cdot \vec{M_s} + \vec{\rho_w}^T \cdot \left(\vec{M_w} + D \cdot \vec{\rho_s} + \frac{1}{2} Q \cdot \vec{\rho_w} \right) =$$

$$= W_0 + \underbrace{\vec{\rho_s}^T \cdot \vec{M_s} + \vec{\rho_w}^T \cdot \vec{M_w}}_{\text{monopole terms}} + \underbrace{\vec{\rho_w}^T \cdot D \cdot \vec{\rho_s}}_{\text{Dipole term}} + \underbrace{\frac{1}{2} \vec{\rho_w}^T \cdot Q \cdot \vec{\rho_w}}_{\text{Quadrupole term}} + \frac{1}{2} \vec{\rho_s}^T \cdot Q_2 \cdot \vec{\rho_s} + \dots$$

↳ less relevant

where W_0, M_s, M_w, D and Q are dependent on z

Recovering the wake functions (until 1st order on $\vec{\rho_s}$ and $\vec{\rho_w}$)

- $W_{\parallel} = \frac{\partial W}{\partial z} \cong W_0' + \vec{\rho_s}^T \cdot \vec{M_s}' + \vec{\rho_w}^T \cdot \vec{M_w}' + \dots$

  * ↳ usually the only term considered for $W_{\parallel}$

  → dominates the expansion ~ usually referred to as the longitudinal wake

  → responsible for longitudinal effects:
  - avg. energy loss
  - potential-well distortion
  - bunch-lengthening / shortening
  - coupled bunch oscillations

- $\vec{W_\perp} = \vec{\nabla}_{w,\perp} W \cong \vec{M_w} + D \cdot \vec{\rho_s} + Q \cdot \vec{\rho_w}$

☆ $\vec{M_s}$ and $\vec{M_w}$ ~ usually referred to as monopolar wakes → effect independent of the transv. displacement of the source and witn. particles

↳ change in the closed orbit as a func. of the current ~ does not lead to instabilities

→ in most practical cases these terms are zero due to symmetries in the vacuum chamber

★ $D = \begin{pmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{pmatrix}$ ~ D_{ij}'s are referred to as <u>dipolar wakes</u> ~ as they are originated from an offset of the source particle from the chamber's center, and generate a wake that is "dipole-like", in the sense that it <u>doesn't depend</u> on the coord.'s x and y of the witness particle!

↳ D_{11} ~ horizontal dipolar wake

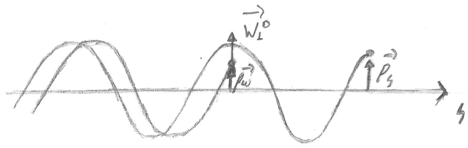
↳ D_{22} ~ vertical dipolar wake

Generates a wake:

$\vec{W_\perp^D} = D \cdot \vec{\rho_s}$, which is thus independent from $\vec{\rho_w}$, just like in the case of a mag. dipole

↳ Responsible for the transverse oscillations that cause coupled bunch oscillations, emittance deterioration and even beam loss

* These terms are particularly dangerous as they describe a mechanism where oscillations of the source particle can induce oscillations on the witness particle, which always leads to a <u>resonant effect</u> as both particles naturally already oscillate with very similar frequencies (equiv. to saying they have approx. the same tune)



→ $D_{12} = D_{21}$ is usually zero due to symmetries in the vacuum chamber, and even when it is not, its effects are not significant as the hor. and vert. tunes are very different, and thus no resonant effects appear - oscillations in one plane do not transfer to oscillations in the other.

★ Q_{11} and Q_{22} ~ Detuning or Quadrupolar wakes

$\vec{W_\perp^Q} = Q \cdot \vec{\rho_w}$ → Wake/kick whose strength depends on the transv. coordinates of the witness particle itself → Just like in a quadrupole magnet

→ generates a focusing effect

Furthermore, if $v \approx c$, $Q_{22} = -Q_{11}$, producing opposite focalizations on x and y, precisely as the mag. quad. does

~ $Q_{12} = Q_{21}$ usually $= 0$ from symmetry considerations

Further understanding the symmetry implications and simplifications that become evident from the wake potential $\left(W(\vec{\rho}_s, \vec{\rho}_w, z)\right)$ formalism (highlighting its power):

Again,

$$W(\vec{\rho}_s, \vec{\rho}_w, z) = W_0 + \vec{\rho}_s^{\,T} \cdot \vec{M}_s + \vec{\rho}_w^{\,T} \cdot \vec{M}_w + \vec{\rho}_w^{\,T} \cdot D \cdot \vec{\rho}_s + \frac{1}{2} \vec{\rho}_w^{\,T} \cdot Q \cdot \vec{\rho}_w$$

Now, let's consider a chamber with <u>mirror symmetry</u> (e.g. rectangular or elliptical)
($x \to -x$ and/or $y \to -y$ do not change W)

Take for instance $x \to -x$: (exception example: radiation masks)

$$W(-x_s, y_s, -x_w, y_w, z) = W(+x_s, y_s, +x_w, y_w, z)$$

- $$(-x_s \;\; y_s) \cdot \begin{pmatrix} M_{sx} \\ M_{sy} \end{pmatrix} = (x_s \;\; y_s) \cdot \begin{pmatrix} M_{sx} \\ M_{sy} \end{pmatrix} \Rightarrow -M_{sx} = M_{sx} \Rightarrow M_{sx} = 0$$

- Analogously for $\vec{\rho}_w^{\,T} \cdot \vec{M}_w$, we get $M_{wx} = 0$

* Doing the same but with $y \to -y$, we get: $M_{sy} = 0$, $M_{wy} = 0$ *

- $$(-x_w \;\; y_w) \cdot \begin{pmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{pmatrix} \cdot \begin{pmatrix} -x_s \\ y_s \end{pmatrix} = (-x_w \;\; y_w) \cdot \begin{pmatrix} -D_{11} x_s + D_{12} y_s \\ -D_{21} x_s + D_{22} y_s \end{pmatrix} = D_{11} x_w x_s - D_{12} x_w y_s - D_{21} x_s y_w + D_{22} y_s y_w,$$

while $$(x_w \;\; y_w) \cdot D \cdot \begin{pmatrix} x_s \\ y_s \end{pmatrix} = D_{11} x_s x_w + D_{12} x_s y_w + D_{21} x_w y_s + D_{22} y_s y_w,$$

and thus, their equality for any x_s, y_s, x_w, y_w demands:

$$\begin{cases} -D_{12} = D_{12} \Rightarrow D_{12} = 0 \\ -D_{21} = D_{21} \Rightarrow D_{21} = 0 \end{cases}$$

- Proceeding analogously for $\vec{\rho}_w^{\,T} \cdot Q \cdot \vec{\rho}_w$, we find $Q_{12} = Q_{21} = 0$

$\Rightarrow$ Non-diagonal terms of D and Q are all zero!

Furthermore, since W is an harmonic function on x_w and y_w for ultra-relativistic electrons,

$\nabla^2_{w,\perp} W = \dfrac{\partial^2 W}{\partial x_w^2} + \dfrac{\partial^2 W}{\partial y_w^2} = 0$, and the quadratic terms with x_w and y_w on the W

expansion (which can thus "survive" two "witness-transverse" derivatives) are:

$2 \cdot W^{Quad} = \vec{\rho}_w^{\,T} \cdot Q \cdot \vec{\rho}_w = Q_{11}\, x_w^2 + 2\, Q_{12}\, x_w\, y_w + Q_{22}\, y_w^2$, we have:

$$\frac{\partial^2 W}{\partial x_w^2} + \frac{\partial^2 W}{\partial y_w^2} = Q_{11} + Q_{22} = 0 \Rightarrow \boxed{Q_{11} = -Q_{22}}$$

And thus for a vacuum chamber w/ mirror symmetry containing ultra-rel. electrons:

- $\vec{M}_s = \vec{M}_w = \vec{0}$
- $D = \begin{pmatrix} D_{11} & 0 \\ 0 & D_{22} \end{pmatrix}$;
- $Q = \begin{pmatrix} -Q_{22} & 0 \\ 0 & Q_{22} \end{pmatrix}$

and W's expansion simplifies to:

$$W(\vec{\rho}_s, \vec{\rho}_w, z) = W_0 + D_{11}\, x_w\, x_s + D_{22}\, y_w\, y_s - \frac{1}{2} Q_{22}\, x_w^2 + \frac{1}{2} Q_{22}\, y_w^2 \quad //$$

W_0, D_{11}, D_{22} and Q_{22} all functions of z.

This enables recovering the recurrent wake functions found in the literature:

defining $W_x^D \triangleq D_{11}$, $W_y^D = D_{22}$, and $W^Q = Q_{22}$, we get:

$$\begin{cases} \bullet\ W_{\parallel}(z) = \dfrac{\partial W}{\partial z} = W_0'(z) \\ \bullet\ W_x(x_s, x_w, z) = W_x^D(z) \cdot x_s - W^Q(z) \cdot x_w \\ \bullet\ W_y(y_s, y_w, z) = W_y^D(z) \cdot y_s + W^Q(z) \cdot y_w \quad // \end{cases}$$

To determine such terms, one can first do a simulation with $x_s \neq 0$, $x_w = 0$ to find W_x^D ($= W_x / x_s$), and then perform another sim. w/ $x_s = 0$, $x_w \neq 0$ to get W^Q ($= -W_x / x_w$)!

Lastly, if the vacuum chamber presents not only mirror symmetry, but also rotation symmetry – i.e., cilindrical symmetry – W must also satisfy:
(exception example: undulators)

$$W(R\cdot\vec{\rho_s},\, R\cdot\vec{\rho_w},\, z) = W(\vec{\rho_s},\, \vec{\rho_w};\, z),$$

where R is a rotation matrix $\left(\text{\#}\ R = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \right)$

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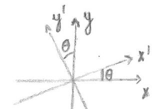

$$x' = x\cos\theta + y\sin\theta$$
$$y' = -x\sin\theta + y\cos\theta$$
$$\Rightarrow \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Plugging into the wake expansion:

$$W(R\cdot\vec{\rho_s},\, R\cdot\vec{\rho_w},\, z) = W_0' + \vec{\rho_w}^{\,T}\, R^T\cdot D\cdot R\, \vec{\rho_s} + \vec{\rho_w}^{\,T}\, R^T\cdot Q\cdot R\, \vec{\rho_w}$$

And thus we need:

- $$R^T\cdot D\cdot R = D \Rightarrow \begin{pmatrix} \cos\theta & -\sin\theta \\ +\sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} D_{11} & 0 \\ 0 & D_{22} \end{pmatrix}\begin{pmatrix} \cos\theta & +\sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} =$$

$$= \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} D_{11}\cos\theta & D_{11}\sin\theta \\ -D_{22}\sin\theta & D_{22}\cos\theta \end{pmatrix} = \begin{pmatrix} D_{11}\cos^2\theta + D_{22}\sin^2\theta & (D_{11}-D_{22})\sin\theta\cos\theta \\ (D_{11}-D_{22})\sin\theta\cos\theta & D_{11}\sin^2\theta + D_{22}\cos^2\theta \end{pmatrix} = \begin{pmatrix} D_{11} & 0 \\ 0 & D_{22} \end{pmatrix}$$

which can only be true if: $D_{11} = D_{22}$, i.e., $W_x^D = W_y^D = W^D$

- Analogously, $R^T\cdot Q\cdot R = Q \Rightarrow Q_{11} = Q_{22}$, but from $\nabla_\perp^2 W = 0$ we have $Q_{11} = -Q_{22}$, leaving as the only option: $Q_{11} = Q_{22} = 0 \Rightarrow W^Q = 0 \Rightarrow$ No quadrupolar wake!

$\Rightarrow$ Therefore, for a perfectly round chamber, $\vec{M_s} = \vec{M_w} = 0$; $D = \begin{pmatrix} W^D & 0 \\ 0 & W^D \end{pmatrix} = W^D\cdot \mathbb{I}$; $Q = 0$, and the wakes become:

- $W_{\parallel}(z) = W_0'(z)$
- $W_x(x_s, z) = W^D\cdot x_s$
- $W_y(y_s, z) = W^D\cdot y_s$

$$\left.\begin{matrix} W_x(x_s,z) = W^D\cdot x_s \\ W_y(y_s,z) = W^D\cdot y_s \end{matrix}\right\} \vec{W_\perp} = W^D\cdot\vec{\rho_s}$$

Thus, in short:

- **General case:**
  - $W_{\parallel}(z) = W_0'(z)$
  - $\vec{W_\perp}(\vec{\rho_s}, \vec{\rho_w}, z) = \vec{M_w}(z) + D\cdot\vec{\rho_s} + Q\cdot\vec{\rho_w}$

- **Mirror - Symmetry**
  - $W_{\parallel}(z) = W_0'(z)$
  - $W_x(x_s, x_w, z) = W_x^D(z)\cdot x_s - W^Q(z)x_w$
  - $W_y(y_s, y_w, z) = W_y^D(z)\cdot y_s + W^Q(z)\,y_w$

- **Axisymmetry**
  - $W_{\parallel}(z) = W_0'(z)$
  - $W_x(x_s, z) = W^D(z)\cdot x_s$
  - $W_y(y_s, z) = W^D(z)\cdot y_s$

1.7 - Properties of the Wakes

* The next sections are based on the books on collective effects from A. W. Chao; K. Y. Ng; and S. Khan.

As discussed previously, for ultra-relativistic particles the wakes satisfy the "Causality Principle", as they travel at the same speed of the particles, c, and thus cannot "catch up" with a particle for which $z \leq 0$. In terms of the wake formalism, this means:

$$W(z \leq 0) = 0 \quad \Rightarrow \quad \begin{cases} \bullet\ \vec{W_\perp}(z) = \vec{\nabla}_{w,\perp} W = 0\,, & \text{for } z \leq 0 \\ \bullet\ W_{\parallel}(z) = \dfrac{\partial}{\partial z} W(z) = 0\,, & \text{for } z \leq 0 \end{cases} \quad \sim \text{considering point particles} \quad \text{(i)}$$

Furthermore, the longitudinal and transverse wakes also satisfy some additional properties, which can be essential to verify if results from simulations are physically plausible/valid. Let's draw realistic $W_{\parallel}$, $W_\perp$ profiles (cylindrical symmetry for simplification, $\vec{W_\perp} = W_\perp\,\hat{\rho}$), and then advance to the properties that explain them!
↳ dipolar wake